



CSE461

Introduction to Robotics Lab

Real-World Implication

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1. **Introduction:** It takes a good amount of time to locate particular books in a big library. A reader has to consult many shelves before finding the appropriate shelf, and librarians devote a portion of their working day to retrieval requests. This is a problem for those who cannot access the high or distant shelves, such as elderly readers and people with reduced mobility. This is done by our Autonomous Library Book Retrieval Robot, called LibBot. It is programmed to pick up a book from a library shelf after identifying it by reading the RFID tags attached to it, and put it down at a collection point, after it is taken from the shelf, without having to move on a fixed library path. Sensors, actuators and an embedded controller help to operate this robot autonomously. This is a small scale prototype of a small library, but the principles will work for a larger library too.
2. **Purpose:** LibBot is about automating the book search process. The robot goes and gets the book, instead of a librarian or reader walking the aisles. This saves time in searching for an item and allows staff to work on other tasks. Another reason for this project is the accessibility. A reader with a disability such as a wheelchair, recovering from an injury, or simply unable to reach a higher shelf can request a book and pick it up without having to drag him or herself across the entire room. Constructing this project involves using autonomous navigation, sensor fusion, motor control, object manipulation and logic to determine what to do next. It is beneficial to learn these parts in operation rather than on paper, working on a real task.
3. **Societal implications:** A library is designed to provide equal access to everything that it contains. Automated retrieval makes it easier for persons who find it difficult to use aisles physically. A person with a healing injury or in a wheelchair can access a book without having to stand on a shelf, or crawl down an awkwardly angled stack of books to retrieve it. A librarian usually works at the front desk of the library, retrieving books, shelving, cataloging and answering questions. The idea is to give chores to a robot to free up time for tasks that require a human, such as assisting with research or directing a reader to the appropriate source. The robot is not sent as a replacement for the human. Instead, it eliminates the repetitive work for them. A robot that is working in a library, and not too loud, is also a way for the people out there to adjust themselves to the concept.
4. **Health implications:** Most of the time, a librarian has to stand on the floor while bending down to access the lower shelves, on a step stool to get to the high shelves, and carry books on shoulder. This can be a cause of tired legs and an odd strain when done throughout the day. Getting the routine fetches to the robot is a bit of that weight off their shoulders.

That same support is provided to someone with injury or to someone older and with mobility issues. They do not have to walk across the room to get a book if they want it,

and they can take it home without having to return it to the school library. The robot must not be constructed to create difficulties for humans. A robot should always go at a slow but constant pace and should not turn quickly or sprint through traffic too close to individuals. It needs to be checked regularly as it could be dangerous due to worn wheels, a loose part or a malfunctioning sensor.

5. **Safety implications:** The most critical safety issue in this project is the sharing of the floor space with people with an autonomous machine. The robot must be able to detect readers or staff moving between the shelves without warning. Moreover, the ultrasonic sensors are alert to objects in their way. If something or someone approaches within a certain distance, it then slows or stops before moving on. It will be kept on course by line-following sensors and RFID markers to prevent it from straying into the open floor.

If the gripper of the robot is sufficiently strong to prevent a book from falling out. But it cannot be too strong to damage the paper of its cover. A book will not be dropped if the gripper of the robot has enough force to stop a book from falling out. The strength, however, should not be too great so as to do damage to the paper of the cover. The robot will not move away from the pick up point without a book in the grip as a limit switch in the gripper verifies that a book is picked up. If something goes wrong, an emergency stop (e-stop) should also be provided to stop all movement at one time which is at the disposal of the librarian. Neat cable management and the battery protection can reduce electrical risk. Regular servicing of the motors, sensors and batteries ensure the robot runs safely everyday.

6. **Legal implications:** A robot in the library is not regular equipment. So if LibBot accidentally knocks over a person, a book, or a shelf, there should be someone to be accountable, probably the university, the lab that made LibBot, or the person who was in charge of the LibBot at the time. For a machine that moves on its own, insurance and liability provisions of library space would have to be revised. Then there is data. Requests for books, if they're recorded, together with those that requested them, would seem like personal borrowing history, and libraries already consider that personal. Any such system would have to comply with any data protection requirements that apply to it and old records would not be retained beyond their usefulness. But there's a smaller legal issue with the RFID tags as well, they could technically be scanned from a distance by anyone who has the right reader to determine what book is where. That would require some thought for a full size deployment, although it may not be a concern for a small prototype. Lastly, if the robot is ever sold or licensed as a product, it would need to conform to the same safety and equipment requirements which apply to other service robots.

7. **Cultural implications:** Libraries are not just places to store books, they are quiet, shared spaces, where people know how to move and where a robot doesn't alter that space. It can be a futuristic idea for some, especially those who are tech-savvy and automation friendly like students. For some patrons, particularly older patrons or those who have frequented the library only to relax and enjoy the quiet and human touch, a rolling robot may seem too much or intrusive. Not to mention the trust factor. People must be assured the robot will not hurt their books, won't be in their way, and will not replace the librarians they know for assistance. It would be very important to introduce it slowly and to make it very noticeable and predictable in its movement, if it was introduced in this way. More broadly, this project is also the beginning of a larger trend. As automation gradually finds a home in society, not just in factories and warehouses, but in a variety of other public spaces, in this case, the library.
8. **Conclusion:** Actually, it is a very common problem that LibBot has been created to solve, spending hours of time in the library searching for a single book. LibBot has covered many different areas to solve the problem, the motors and sensors and the code. It's also about making a library comfortable to use for those who have difficulty moving around, less burden on librarians, a safe environment for a robot moving around a library, and planning ahead for legal and social issues of a robot in a public place. The ideas, how to navigate, how to identify the objects and how to handle them would be the same in a real library, except that they would be larger. In the end, a project like this is a reminder that even a simple automation idea, when followed through from start to finish, has implications in the realms of technology, people and responsibility all at once.